

This Time Is Not So Different: Volatility Regimes In The Cryptocurrency Market

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Date of submission: June 8, 2026

Abstract

In conventional markets, volatility admits a decomposition into a systematic component and an idiosyncratic residual. No such decomposition yet exists for a contemporary cryptocurrency universe at the individual-asset level. We build it by applying to a panel of 38 cryptoassets (91% of market capitalization, October 2018 to February 2026) the asymptotic principal components factor model of Connor, Korajczyk and Linton (2006) and the residual COVOL framework of Engle and Campos-Martins (2023), and we obtain a daily per-asset decomposition into common and idiosyncratic components. The cross-section alternates between two regimes that a Markov-switching regression separates sharply: systemic stress with a common share above 55%, and speculative below 30%. Within that architecture, the two largest single-day shocks in the sample do not occur during the crypto-native failures, but at the SEC v. Ripple ruling and the spot Bitcoin ETF approval; LUNA / Terra is the only episode with sustained elevation. Macro-financial transmission turns out to be selective: crypto-native sentiment, captured by the Fear & Greed index, reaches the idiosyncratic volatility of nearly the entire cross-section, while the VIX, the EPU, the NFCI, the GPR, the dollar and oil add no explanatory power at the daily frequency. Applied to 38 S&P 500 stocks and 11 SPDR sector ETFs under the same procedure, the exercise yields an indistinguishable regime architecture; the differential concentrates in the upper tail and in crypto-native events. In the space of volatility, this time is not so different.

Keywords: Cryptocurrencies; Co-volatility; COVOL; Factor model; GARCH

JEL classification: G12, G15, G17, G28

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Author contributions (CRediT taxonomy). Both authors contributed to: conceptualisation, methodology, formal analysis, software, validation, data curation, writing (original draft), writing (review and editing). Both authors read and approved the final manuscript.

Declarations

Funding. The authors declare that no specific funding was received for this work.

Competing interests. The authors have no competing interests, financial or otherwise, to disclose in relation to the content of this manuscript.

Data availability. All data used in this study are obtained from publicly available sources: cryptocurrency log-returns from CoinMarketCap; the VIX, the U.S. Dollar Index and WTI crude prices from the Federal Reserve Bank of St. Louis (FRED); the Economic Policy Uncertainty (EPU) index from policyuncertainty.com; the Geopolitical Risk (GPR) index from matteoiacoviello.com; the National Financial Conditions Index (NFCI) from the Federal Reserve Bank of Chicago; the Fear & Greed index from alternative.me. The full universe with tickers and market shares is reported in the Online Appendix.

Code availability. Replication code (Python) for the full APC + COVOL pipeline, the Markov-switching identification, the bootstrap inference and all tables and figures is available from the corresponding author upon reasonable request.

Ethical approval. Not applicable; the study uses only publicly available financial market data and does not involve human or animal subjects.

Use of generative AI. The authors used generative AI tools to assist with language editing of selected passages of the manuscript. The authors take full responsibility for the content of the publication.

Acknowledgements

The authors thank seminar participants at Universidad Diego Portales for helpful comments on earlier drafts of this work.